

Mand.ro Mark 7

Communication Protocol

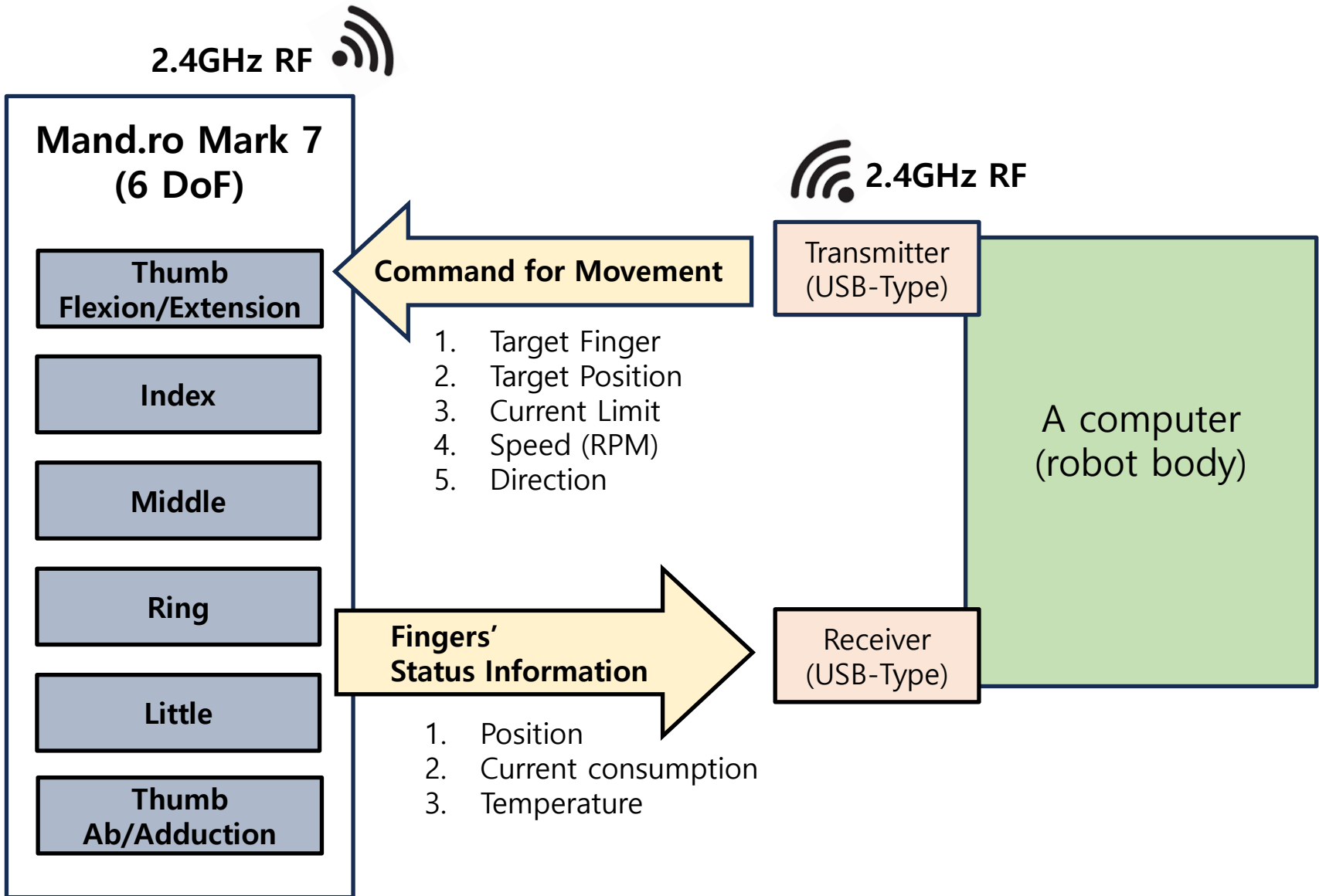
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Sangho Yi / Mand.ro

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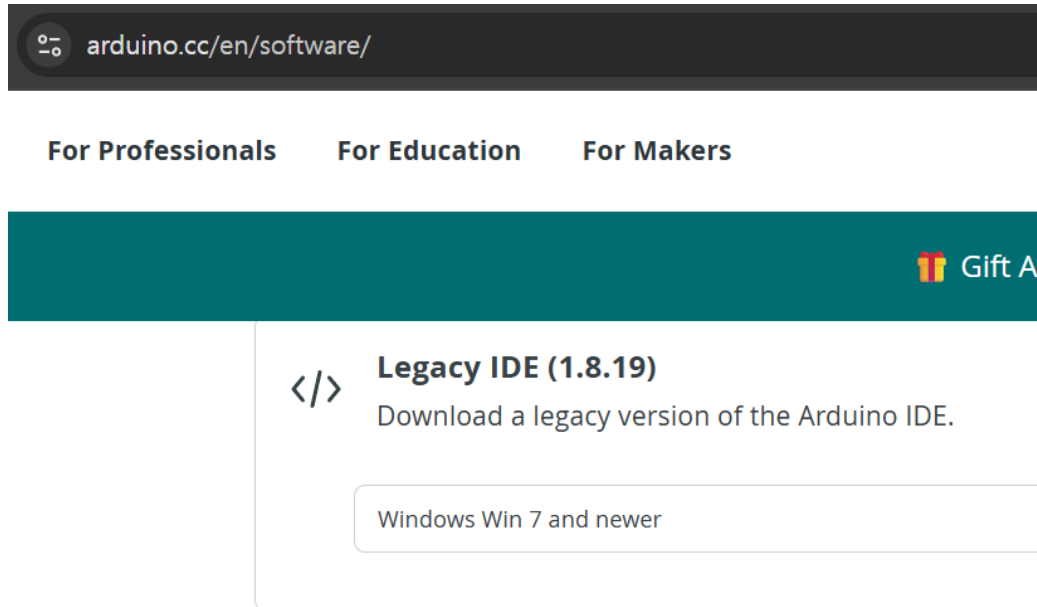
- Mand.ro Mark 7
- Note: Test Program – Comport Master

Mand.ro Mark 7

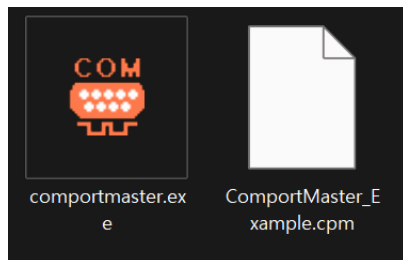


Prerequisites

- Install Arduino IDE (1.8.19 is recommended version)
 - This is required to install a USB-Serial Driver for Arduino-compatible boards.
 - Note that we use Arduino Micro board for our USB Type-A dongle for wireless communication.



- Install Comport Master program
 - This is a third-party program for easier Serial communication and testing.



PC to Hand Command Protocol (via Transmitter Dongle)

- Insert a USB Type-A Dongle into a PC, and Establish a Serial (COM port) Communication
 - Port Name: Arduino Micro (COM xx)
 - Baud Rate: 115200 bps (8-N-1)
- Tx Packet Length: 11 bytes
 - Hand:1, Finger: 1, Speed: 1, Current: 1, Position: 6, Direction: 1

Sequence	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
Description	Hand	Finger	Speed	Current	Position						Direction
					Thumb Flexion (High 4bit)	Index Flexion (Low 4bit)	Middle Flexion (High 4bit)	Ring Flexion (Low 4bit)	Little Flexion (High 4bit)	Thumb Abduction (Low 4bit)	
Data	0xFD (Left)	0x000000 ... 0x111111	0: skip setting 10: 6K RPM 20: 7K RPM 50: 11K RPM 100: 15K RPM 200: 25K RPM 250: 30K RPM	0: skip setting 10: 630 mA 20: 660 mA 50: 750 mA 100: 900 mA 200: 1200 mA 250: 1350 mA	0: skip setting 1: -19 Steps 10: -1 Steps 50: 79 Steps 100: 179 Steps 200: 379 Steps 250: 429 Steps						0, 1, 2, 3
	0xFE (Right)										
	0xFF (Both)										
Note	Hand Selector	Bitwise Selection 1: Thumb 2: Index 4: Middle 8: Ring 16: Little 32: Thumb Abd	RPM/200 (30 ~ 150) 50 means 10K RPM	Calculation: 600 + Data * 3	Calculation: -21 + Data * 2 * Due to the counter shifts from hall-effect sensor encoder, we provide a minus position to full-release.						0: Idle 1: Forward 2: Reverse 3: Reset
Extra Note					Ignored by 'zero' value						

Important Note for Absolute Position Control

- The absolute position is calculated from the initial finger position. (fully released)
- However, **the hall-effect sensors in the motor may have slips**, so it is recommended to reset its position counter when the hand is fully released while using.
- With the absolute position parameter, the 'forward direction' is the only one command to control the position. For example, if you make the following commands sequentially by time:
 - 1) Forward, Position Set as 100 -> Goes from zero to 100 position (grasping movement)
 - 2) Forward, Position Set as 50 -> Goes from 100 to 50 position (reverse action)
 - 3) Forward, Position Set as 200 -> Goes from 50 to 200 position (grasping movement)
 - 4) Forward, Position Set as **-10** -> Goes from 200 to **-10** position (reverse action)
 - * -10 is not reachable position, but the position counter has slips, so it is better to make below zero position setting for full releasing fingers
- **Note that setting zero (0) as a position value means it does not change the target position previously given to the finger.**
- The position parameter can be **reset to** zero value by **reset command (direction as 3).**

Hand to PC Command Protocol (via Rx Dongle)

- With the same USB Type-A dongle, you can receive feedback via RF 2.4 GHz wireless communication.
 - Port Name: Arduino Micro (COM xx)
 - Baud Rate: 115200 bps (8-N-1)
- Data Format (plain text string, divided by 'comma' / , /)
 - L | R (Left or Right)
 - Position, Current, Temperature (Thumb Flexion)
 - Position, Current, Temperature (Index Flexion)
 - Position, Current, Temperature (Middle Flexion)
 - Position, Current, Temperature (Ring Flexion)
 - Position, Current, Temperature (Little Flexion)
 - Position, Current, Temperature (Thumb Abduction)

Note: Test Program: Comport Master (third-party program)

ComPortMaster - Withrobot Co.

Port Config

Device: COM52

Baudrate: 115200

Data bits: 8

Stop bits: 1

Parity: None

Open port Close port

Port Control & Status

☒ RTS ☒ DTR

☐ CTS ☐ DSR ☐ RI ☐ CD

Update Status

Tools & Options

☐ Use Extended SLIP

☒ LF ☐ CR/LF

Import Export

Show Device Manager

Quick Send

0 1 2 3 4 5 6 7 8 9 A B C D E F Space CR/LF ☐ ASCII ☐

Send

☐ FF 3F B0 A0 50 60 70 80 90 A0 01 ☒ Hex ☐ Send

☐ FF 3F B0 A0 01 01 01 01 01 01 01 ☒ Hex ☐ Send

☐ ☒ Hex ☐ Send

☐ FF 3F B0 A0 00 00 00 00 00 00 01 ☒ Hex ☐ Send

☐ ☒ Hex ☐ Send

☐ ☒ Hex ☐ Send

☐ ☒ Hex ☐ Send

☐ ☒ Hex ☐ Send

Send Multiple : 0 / 1 ☐ Interval 100 ms Cancel

Recv

☐ Decode SLIP ☒ Auto CR/LF ☐ Handle CR/LF Start Capture ASCII Clear

0....|....1....|....2....|....3....|....4....|....5....|....6....|....7....|....8

-> [LR] F:111111 S:35200 C:1140 P:+139 +171 +203 +235 +267 +299 D:1

-> [LR] F:111111 S:35200 C:1140 P:+139 +171 +203 +235 +267 +299 D:1

-> [LR] F:111111 S:35200 C:1140 P:+139 +171 +203 +235 +267 +299 D:1

-> [LR] F:111111 S:35200 C:1140 P:+139 +171 +203 +235 +267 +299 D:1

-> [LR] F:111111 S:35200 C:1140 P:+139 +171 +203 +235 +267 +299 D:1

